

LINE REPRESENTATION

TO/FROM DESCRIPTION + TAG No.

XX.XX.XX.XXXX

OPC

NO POCKET

NO LOW POINT LINE

SLOPE

1:X

SLOPE

SLOPE 1:300

CLOSED DRAIN LINES (MAIN HEADERS)

SLOPE 1:100

CLOSED DRAIN LINES (SUB HEADERS)

SLOPE

LINE TO SLOPE DOWN (UNDEFINED)

F.D.

FREE DRAINING

DATA BUS

SPEC. BREAK

HEAT TRACING (ELECTRICAL)

PERSONAL PROTECTION PP

FENCE

VENDOR TR-CCC

REVISION CLOUD

SCOPE CLOUD

IN LINE EQUIPMENT

HOSE CONNECTION

PLUG (SCREWED)

BLIND FLANGE

CAP (BUTT WELDED)

SPECTACLE BLIND (OPEN)

SPECTACLE BLIND (CLOSED)

FLOW STRAIGHTENING VANES

VACUUM BREAKER OR BREATHER VALVE

FILTER (BASIC SYMBOL)

PULSE JET FILTER

STRAINER "T" TYPE

STRAINER "Y" TYPE

CONICAL STRAINER

REDUCER

FLANGE OR HUB

FLEXIBLE CONNECTION

FLAME ARRESTOR

STEAM TRAP

DEMISTER

WATER DISPOSAL WELL (FOR PFD)

SILENCER

MONITOR

SAMPLE CONNECTION

STRAINER BASIC SYMBOL

FIRE HYDRANT

VALVE WITH BLIND

FIRE/GAS DETECTOR

BARRED TEE

FINNED TUBE AIR COOLER

STATIC MIXER

INSULATING FLANGE

FIRE WATER MONITOR (120m3/h)

TRAP DRAIN

FLOW DETECTOR SWITCH

FLOUR-WAY HYDRANT WITH WATER MONITOR (FOR PIPING DRGS ONLY)

REMOVABLE SPOOL

BLIND DISC

SPACER (RING SPACER)

SIGHT GLASS

MECHANICAL PIG SIGNALER

SP TAG

BASKET STRAINER

MONOBLOCK INSULATING JOINT

TUNDISH DRAIN OR OPEN DRAIN

MONOBLOCK ASSEMBLY WITH DOUBLE BLOCK AND BLEED

TIE-IN TO EXISTING LINE OR EQUIPMENT

CHEMICAL INJECTION POINT

CORROSION COUPON

CORROSION PROBE

CORROSION MONITOR (ELECTRICAL RESISTANCE)

BIOPROBE

VORTEX BREAKER

VENT WITH BIRD SCREEN

ATMOSPHERIC VENT

ELECTRICAL RESISTANCE PROBE

SPRAY NOZZLE

HYDRANT HOSE BOX

FIRE WATER MONITOR (200m3/h)

PULSATION DAMPENER

U-BEND

CHARCOAL FILTER

HALF OPEN PIPE

CONTROL VALVES

EXTERNAL TAPPING PRESSURE/REDUCING/REGULATING VALVE

DELUGE VALVE

EXTERNAL TAPPING BACK PRESSURE/REDUCING/REGULATING VALVE

THERMAL SAFETY RELIEF VALVE

PRESSURE/REDUCING/REGULATING VALVE SELF CONTAINED

SOLENOID OPERATED VALVE

BACK PRESSURE REDUCING/REGULATING VALVE SELF CONTAINED

VACUUM SAFETY OR RELIEF VALVE

CONTROL VALVE OPEN ON AIR FAIL OR/AND ELECTRICAL FAIL

PRESSURE SAFETY OR RELIEF VALVE OR THERMAL SAFETY VALVE

CONTROL VALVE CLOSED ON AIR FAIL OR/AND ELECTRICAL FAIL

PRESSURE/VACUUM SAFETY OR RELIEF VALVE

CONTROL VALVE WITH HAND WHEEL

SUB-SURFACE SAFETY VALVE

THREE WAYS VALVE

DIFFERENTIAL PRESSURE REDUCING/REGULATING VALVE

DIAPHRAGM OPERATED VALVE

SURFACE SAFETY VALVE

PISTON OPERATED VALVE

GAS ACTUATED ROV

MOTOR OPERATED VALVE

SOLENOID OPERATED VALVE (2 WAY)

SHUTDOWN VALVE

REMOTE OPERATED VALVE

GLOBE VALVE TYPE

BLOWDOWN VALVE

BALL VALVE TYPE

ELECTRONIC WATER DRAIN VALVE

BUTTERFLY VALVE TYPE

ABBREVIATIONS-PROCESS

A/G — ABOVE GROUND

MP — MEDIUM PRESSURE

MPSE — MULTIPHASE FLOW METER

MSM — MULTIPPOINT SELECTOR MANIFOLD

NC — NORMALLY CLOSED

ND — NOMINAL DIAMETER

NNF — NORMALLY NO FLOW

NO — NORMALLY OPEN

NRV — NON-RETURN VALVE

O — OPEN

O/F — OUTER FLOW

OFR — OIL FLOW RATE(BOPD)

PS — PIPE SPOOL

PLC — PROGRAMMABLE LOGIC CONTROLLER

PRV — PRESSURE REGULATING VALVE

PSV — PRESSURE/VACUUM RELIEF VALVE

RDS — REMOTE DEGASSING STATION

RO — RESTRICTING ORIFICE

RS — REMOVABLE SPOOL

RTU — REMOTE TERMINAL UNIT

RV — RELIEF VALVE

SAS — SAHIL/ASAB/SHAH

SC — SAMPLE CONNECTION

Scrd — SCREWED

SDV — SHUTDOWN VALVE

SH — SEQUENCE ALARM HIGH

SL — SEQUENCE ALARM LOW

SO — STEAMING OUT

SP — SET PRESSURE

STP — SEWAGE TREATMENT PACKAGE

SS — SHORT STRING

SSSV — SUB-SURFACE SAFETY VALVE

SSV — SURFACE SAFETY VALVE

SW — SOCKET WELD

TC — THROUGH CONDUIT

TGF — TOTAL GAS FLOW(MMSCF FROM START TEST)

TOF — TOTAL OIL FLOW(BO FROM START TEST)

TSO — TIGHT SHUT OFF(CLASS VI)

TOV — TRIPLE OFFSET VALVE

TOP — TOP OF PIPE

TWF — TOTAL WATER FLOW(BW FROM START TEST)

TW — THERMOWELL

UC — UTILITY CONNECTION

UCP — UNIT CONTROL PANEL

U/F — UNDER FLOW

U/G — UNDER GROUND

V — VENT

VFD — VARIABLE FREQUENCY DRIVE

VSD — VARIABLE SPEED DRIVE

VRC — VAPOUR RECOVERY COMPRESSOR

WAG — WATER ALTERNATING GAS

WC — WATER CUT

WFR — WATER FLOW RATE(BWPD)

WHCP — WELLHEAD CONTROL PANEL

Z — CHEMICALS

MOV — MOTOR OPERATED VALVE

MP — MEDIUM PRESSURE

MPSE — MULTIPHASE FLOW METER

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WHCP — WELLHEAD CONTROL PANEL

Z — CHEMICALS

INSULATION NOMENCLATURE APPLICABLE TO FFD PROJECT :

H — THERMAL HOT

P — PERSONNEL PROTECTION

A — ACOUSTIC

C — THERMAL COLD

T — ELECTRICAL HEAT TRACING

INSULATION NOMENCLATURE APPLICABLE TO SAHIL PHASE 3 PROJECT :

H — FOR HEAT CONSERVATION AND PROCESS TEMPERATURE CONTROL

PP — PERSONNEL PROTECTION

E — ELECTRICAL TRACED LINE AND INSULATED

FP — FOR PROTECTION OF PIPING AND EQUIPMENT UNDER FIRE

AA — ACOUSTIC INSULATION

N — NO INSULATION

EQUIPMENT NUMBERING

AREA CODE

PLANT-AREA CODE

EQUIPMENT CODE

SYSTEM CODE

UNIT No.

SEQUENCE No.

XX - XX - XXXX - XXXX - XX

LINE NUMBERING FOR PIPING

SIZE (INCHES)

SERVICE CODE

SERIAL No.

SPECIFICATION

INSULATION

XX - XX - XXXX - XXXX - X - X

INSULATION NOMENCLATURE APPLICABLE TO FFD PROJECT :

H — THERMAL HOT

P — PERSONNEL PROTECTION

A — ACOUSTIC

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INSULATION NOMENCLATURE APPLICABLE TO SAHIL PHASE 3 PROJECT :

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PP — PERSONNEL PROTECTION

E — ELECTRICAL TRACED LINE AND INSULATED

FP — FOR PROTECTION OF PIPING AND EQUIPMENT UNDER FIRE

AA — ACOUSTIC INSULATION

N — NO INSULATION

LINE NUMBERING FOR PIPELINE

SIZE (INCHES)

SERVICE CODE

SERIAL No.

SPECIFICATION

XX - XX - XXXX - XXXXXX

NOTES

1. FOR ADDITIONAL SYMBOLS/INFORMATIONS AND ALL THE CODES, REFER EP.30.99.90.001.

AS — AIR SUPPLY

ES — ELECTRICAL SUPPLY

GS — GAS SUPPLY

IA — INSTRUMENT AIR

NS — NITROGEN SUPPLY

WS — WATER SUPPLY

2. FOR PIPING CLASSES REFER TO PIPING MATERIAL SPECIFICATION.

3. REFER TO "PIPING STANDARD FOR DESIGN AND CONSTRUCTION 30.99.23.1601".

4. THIS SHUTDOWN SYMBOLGY HAS BEEN SUPERSEDED. HOWEVER IT SHALL REMAIN FOR EXISTING EQUIPMENT. NEW SAS EQUIPMENT SHALL USE SHUTDOWN SYMBOLGY AS NOTE 5.

5. THIS SYMBOLGY SHALL BE USED TO IDENTIFY ESD FUNCTIONS FOR NEW SAS EQUIPMENT. WHEN USED FOR OTHER PLC SYSTEMS, THE SYSTEM IDENTIFICATION SHALL BE ADDED ADJACENT TO THE SYMBOL.

6. THE FF SYMBOL SHALL BE USED TO IDENTIFY FOUNDATION FIELDBUS INSTRUMENTS.

7. ONLY FOR INSTRUMENTS.

8. ALL VALVES SHOWN AT EQUIPMENT FLANGE WILL BE LOCATED FLANGE-TO-FLANGE AT THE NOZZLE, WITH THE SAME NOZZLE ORIENTATION.

9. FOR ESD TRANSMITTERS WILL EXIST A DISCREPANCY ALARM WITH ITS DCS COUNTERPART WHEN APPLIES.

10. THE FOLLOWING ITEMS SHALL BE IDENTIFIED AND TAGGED AS SPECIALITY ITEMS.
ACCESS FITTINGS (INJECTION QUILL), BIRD SCREEN, MECHANICAL INTERLOCK, STATIC MIXER QUICK COUPLING AND HOSE, SIGHT GLASS, FLAME ARRESTOR

HOLDS

REFERENCE DRAWINGS

TITLE

DRG. No.

LEGEND

SCOPE CLOUD FOR PROJECT NO. P16093

REVISION CLOUD FOR PROJECT NO. P16093

CLOUDED PORTION DENOTES :
SCOPE FOR THIS PROJECT (P16093)

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DWG NO. 30-99-08-0101, SHT 1 OF 2, REV 1

LIST OF LINE SERVICE CODES

CODE

SERVICE

AW

Aquifer Water

BD

Blowdown Drain (Automatic Process Drain)

BG

Blanket Gas

BZ

Biocide

CG

CO2

CW

Cooling Water

CZ

Flocculant / Coagulant

D

Drains (General)

DF

Diesel Fuel

DW

Potable Water/Distilled Water

DZ

Demulsifier

FG

Fuel Gas (includes purge, stripping)

FW

Fire Water

FZ

Foam Concentrate

GL

Glycol (TEG:Triethylene Glycol)

HC

Hydrocarbon Condensate

HM

Heating Medium

HO

Hydraulic Oil

HZ

Sodium hypochlorite

IA

Instrument Air

IG

Injection Gas

IW

Injection Water (Surface pump discharge to injection wells)

IZ

Corrosion Inhibitor

LO

Lube Oil

MF

Motor Gasoline Fuel

NN

Nitrogen

OZ

Oxygen Scavenger

PA

Plant Air

PG

Process Gas

PO

Process Oil (all Crude streams)

PW

Produced Water (From Process sources)

R

Relief (General)

RW

Raw Water

SD

Sewage Drain

SG

Seal Gas

SO

Seal Oil

SZ

Scale Preventative (anti-scale)

V

Vent

VG

Vent Gas (Flare)

VT

Vessel Trim

WM

Washing Fluid Media

WW

Wash Water

Z

Chemicals (antifoam, methanol, wax inhibitor, PH control chemical)

ZO

Sludge

INSTRUMENTS

ACTUATOR RESET

ANALYSER ELEMENT (PROBE)

LOCALLY MOUNTED (NOTE 6)

MOUNTED ON MAIN CONTROL ROOM PANEL

MOUNTED ON MAIN CONTROL ROOM BACK OF PANEL

MOUNTED ON LOCAL PANEL

MOUNTED ON BACK OF LOCAL PANEL

DISTRIBUTED CONTROL SYSTEM ITEMS NORMALLY ACCESSIBLE TO THE OPERATOR INDICATOR/CONTROLLER/RECORDER OR ALARM POINTS USUALLY USED TO INDICATE VIDEO DISPLAY

DCS NOT ACCESSIBLE

PLC

PLC NOT ACCESSIBLE

PLC LOCAL PANEL

BOOLEAN LOGIC

HIGH SELECTOR

LOW SELECTOR

INTERLOCK

PILOT LIGHT

HIPPS

SIGNAL FROM TELEMETRY

SIGNAL TO TELEMETRY

TURBINE FLOWMETER

ESD

SHUTDOWN SIGNAL IN (TO INSTR) (NOTE 4)

SHUTDOWN SIGNAL OUT (FROM INSTR) (NOTE 4)

SNUBBER

2oo3 : TWO OUT OF THREE (VOTING SYSTEM)

1oo2 : ONE OUT OF TWO (VOTING SYSTEM)

LOCAL PUSH BUTTON

MANUAL RESET PUSH BUTTON

BUTTERFLY VALVE WITH NATURAL BYPASS

DISPLACEMENT TYPE LEVEL TRANSMITTER

GUIDED WAVE RADAR TYPE LEVEL TRANSMITTER

DIFFERENTIAL PRESSURE TYPE LEVEL TRANSMITTER

DIFFERENTIAL PRESSURE TYPE LEVEL TRANSMITTER (DIAPHRAGM SEAL)

LEVEL GAUGE (REFLEX OR TRANSPARENT)

MAGNETIC LEVEL

FUNCTION AVAILABLE IN SUPERVISOR MONITORING SYSTEM OF ADNOC

DIAPHRAGM

VELOCITY CHECK VALVE

ROTAMETER TYPE FLOW INDICATOR

MAGNETIC FLOW METER

VORTEX FLOW METER

V-CONE

ORIFICE PLATE

ORIFICE PLATE-QUICK CHANGE FITTING

ULTRASONIC FLOWMETER

MASS FLOWMETER

POSITIVE DISPLACEMENT FLOWMETER

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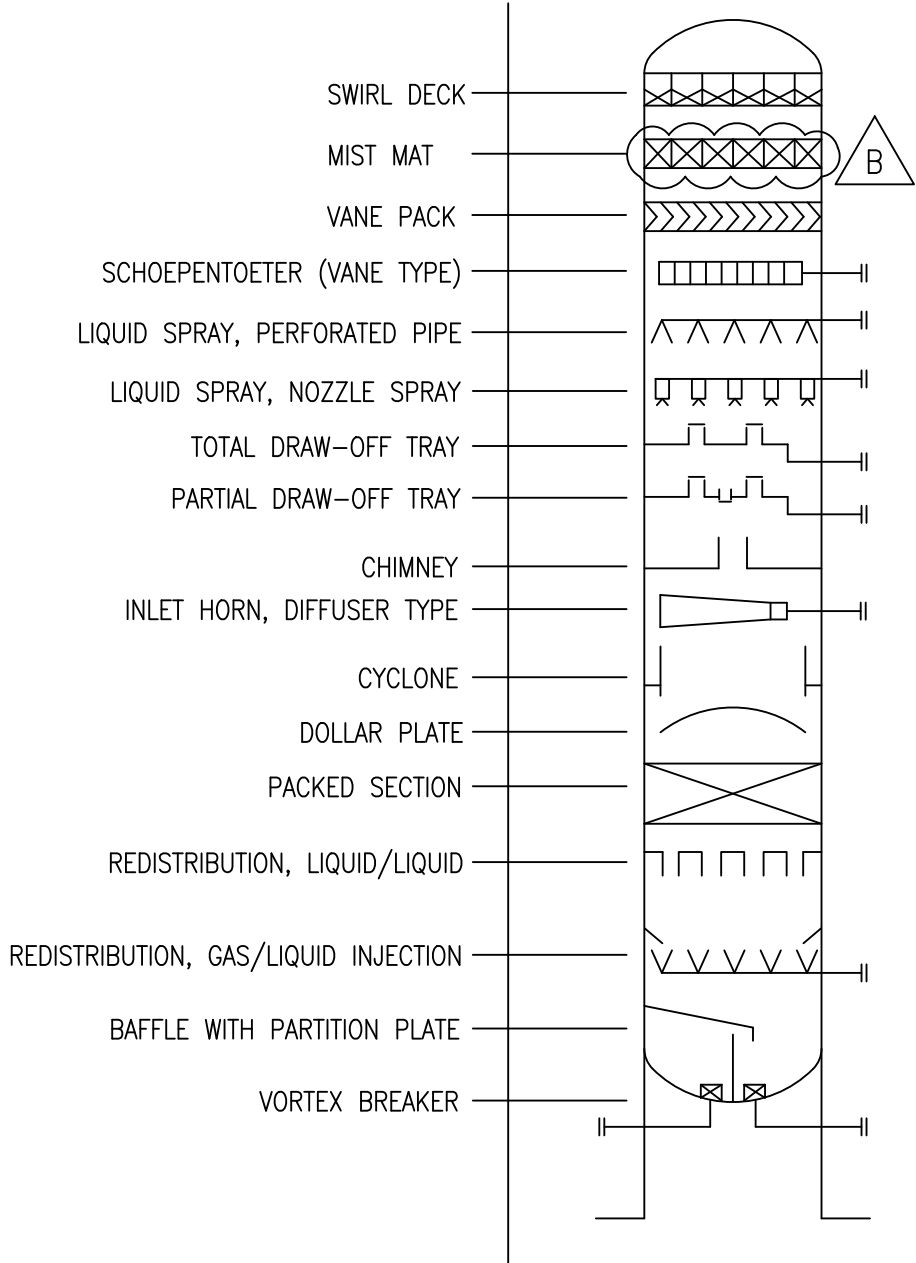
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P16093-30-99-08-1601-1.DWG

ABBREVIATIONS–INSTRUMENTATION

	FIRST LETTER		SUCCEEDING LETTER		
	MEASURED OR INITIATING VARIABLE	MODIFIER	READOUT /PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	ANALYSIS		ALARM		
B	BURNER FLAME		USER'S CHOICE	USER'S CHOICE	USER'S CHOICE
C	CONDUCTIVITY (ELECTRICAL)			CONTROL	
D	DENSITY (MASS) OR SPECIFIC GRAVITY	DIFFERENTIAL			
E	VOLTAGE (EMF)		PRIMARY ELEMENT		
F	FLOWRATE	RATIO (FRACTION) FIRE DETECTION			
G	GAGING (DIMENSIONAL)	GAS DETECTION	GLASS		
H	HAND (MANUALLY INITIATED)	HEAT DETECTION			HIGH
I	CURRENT (ELECTRICAL)		INDICATE		
J	POWER	SCAN			
K	TIME / TIME SCHEDULE			STATION	
L	LEVEL		LIGHT (PILOT)		LOW
M	MOISTURE OR HUMIDITY	MOTOR OPERATED			MIDDLE OR INTERMEDIATE
N	USER'S CHOICE		USER'S CHOICE	USER'S CHOICE	USER'S CHOICE
O	USER'S CHOICE		ORIFICE (RESTRICTION)		
P	PRESSURE OR VACUUM		POINT (TEST CONN.)		
Q	QUANTITY OR EVENT	INTEGRATE OR TOTALIZER			
R	RADIOACTIVITY		RECORD OR PRINT		
S	SPEED OR FREQUENCY	SAFETY SMOKE DETECTION		SWITCH	
T	TEMPERATURE			TRANSMIT	
U	MULTIVARIABLE	RATIO (FRACTION) FIRE DECTION	MULTIFUNCTION	MULTIFUNCTION	MULTIFUNCTION
V	VISCOSITY	VIBRATION		VALVE, DUMPER OR LOUVER	
W	WEIGHT / FORCE		WELL		
X	UNCLASSIFIED		UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
Y	USER'S CHOICE			RELAY OR COMPUTE	
Z	POSITION	ZENER DIODE		DRIVE, ACTUATE OR UNCLASSIFIED FINAL CONTROL ELEMENT	

COLUMN INTERNALS (NOTE 4)



IDENTIFICATION OF EQUIPMENT

LETTER OF GROUP	DESCRIPTION
C	COLUMNS : TRAY COLUMNS PACKED COLUMNS ROTATING DISC CONTACTORS
D	DRYING EQUIPMENT
E	HEAT TRANSFER EQUIPMENT : HEAT EXCHANGERS CONDENSERS COOLERS REBOILERS
F	FAN / BLOWER
J	JET (EJECTORS, INJECTORS AND EDUCTORS)
K	COMPRESSORS, BLOWERS, FANS VACUUM PUMPS, LIQUID RING PUMPS
M	MANIFOLD
P	PUMPS (CENTRIFUGAL, RECIPROCATING, ROTARY)
S	FILTERS
T	TANKS
U	SKID / PACKAGE
V	VESSELS, INCL. PRESSURE STORAGE VESSELS
X	MISCELLANEOUS
RP	PIG RECEIVER
LP	PIG LAUNCHER
SC	SLUG CATCHER

DRIVERS (NOTE-2)

LETTER OF GROUP	DESCRIPTION
M	MOTORS (ELECTRICAL)
D	DIESEL ENGINE

NOTES

- FOR PIPING CLASSES REFER PIPING MATERIAL SPECIFICATION.
- DRIVERS CODES SHALL BE USED IN 2nd, 3rd AND 4th POSITION OF EQUIPMENT CODE TO INDICATE THE SUPPORTING DRIVING FUNCTION OF A MAIN EQUIPMENT.
- FOR ADDITIONAL SYMBOLS / INFORMATION AND ALL CODES REFER TO 30.99.90.0001 REV 7.
- EQUIPMENT SHOWN FOR CLARITY.

HOLDS

REFERENCE DRAWINGS

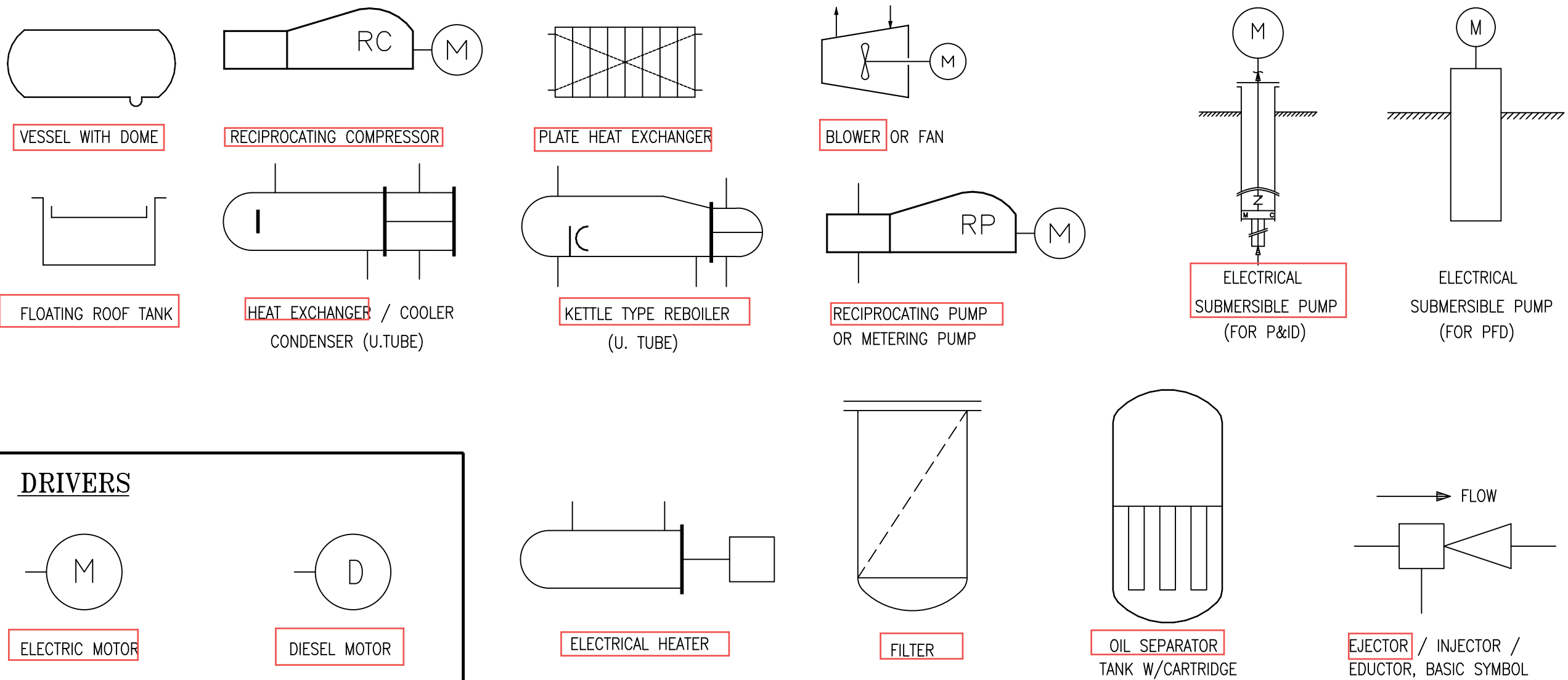
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LEGEND

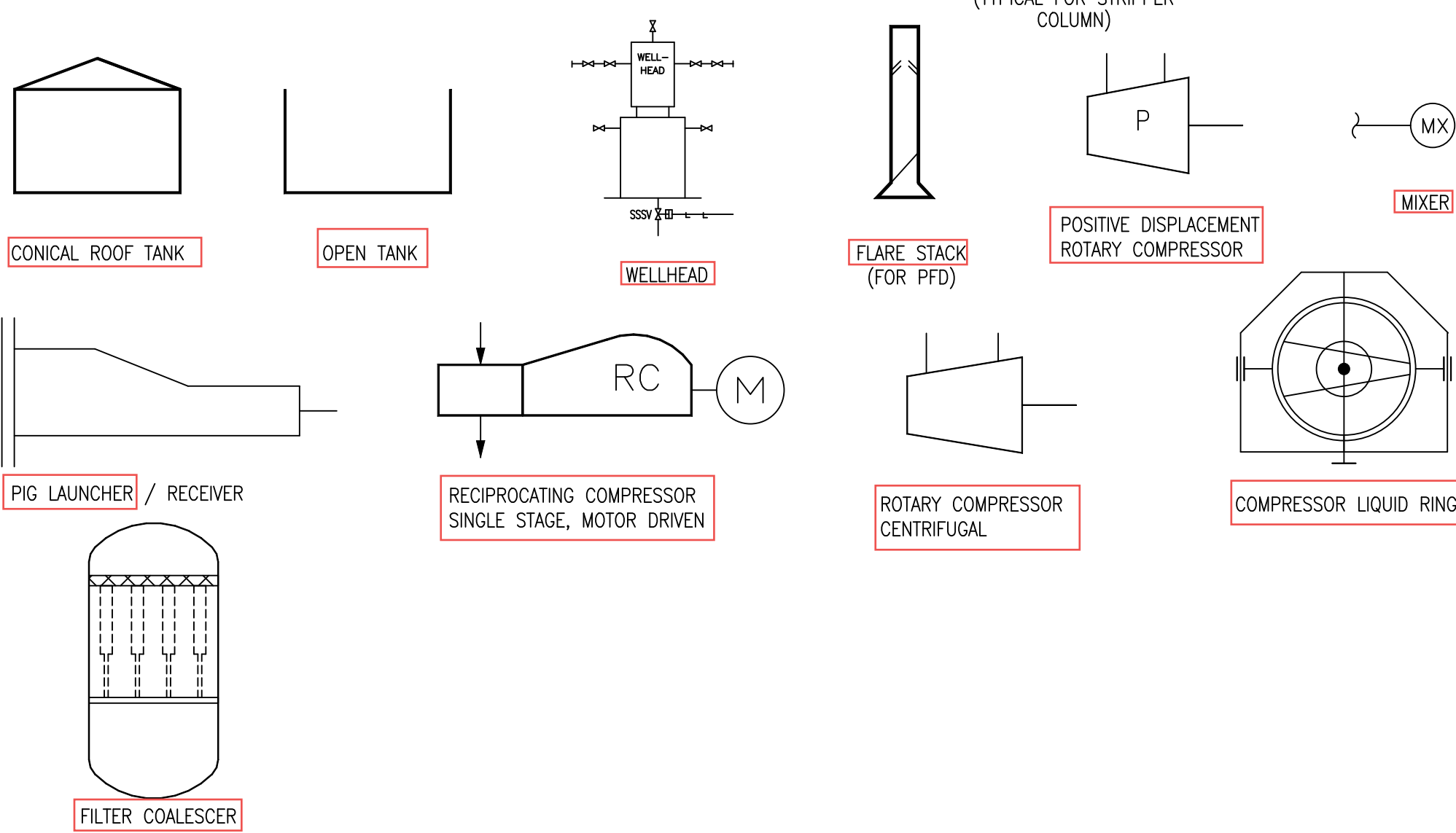
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- REVISION CLOUD FOR PROJECT NO. P16093

CLOUDED PORTION DENOTES :
SCOPE FOR THIS PROJECT (P16093)

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DWG NO. 30-99-08-0101, SHT 2 OF 2, REV 1



DRIVERS



FOR
INFORMATION
ONLY

Standard P&ID Symbols Legend | Industry Standardized P&ID Symbols

Piping and Instrument Diagram Standard Symbols Detailed Documentation provides a standard set of shapes & symbols for documenting P&ID and PFD, including standard shapes of instrument, valves, pump, heating exchanges, mixers, crushers, vessels, compressors, filters, motors and connecting shapes.

Instrument

	Indicator		Computer Indicator		Level Controller		Magnetic		Or Gate
	Behind Control		Programmable Indicator		Pressure Transmitter		Pilot Tube Type Flow		Not Gate
	On Central Control		Displayed Programmable Device		Pressure Recorder		Pilot Tube		Correcting Element
	On Local Control Pane		Computer		Pressure Controller		Wedge Meter		Diamond
	Behind a Local Control		Unit Control Panel		Pressure Indicating		Target Meter		Pressure Gauges
	Indicator 2		Temp Indicator		Pressure Recording		Weir Meter		Thermometers
	Indicator 3		Temp Transmitter		Level Alarm		Ultrasonic Meter		Averaging Pilot Tube
	Indicator 4		Temp Recorder		Flow Element		V-cone Meter		Level Meter
	Indicator 5		Temp Controller		Temperature Element		Venturi Meter		Coriolis Flow Sensor
	Odometer		Flow Indicator		Level Gauge		Quick Change		Flow Nozzle Meter
	Pressure Gauge		Flow Transmitter		Analyzer Transmitter		Turbine Meter		Flume Meter
	Flowmeter		Flow Recorder		Transducer		Rotometer		Manhole
	Thermometer		Flow Controller		Specialty Item		Double		Socket Connection
	Shared Indicator		Level Indicator		Sampler		Steam Traced		Support Bracket
	Displayed Configurable		Pressure Indicator		Straightening Vanes		Venturi		Support Leg
	Shared Indicator		Level Transmitter		Diaphragm Meter		Flowmeter		Support Ring
			Level Recorder		Rotary Meter		Vortex Sensor		Support Skirt
							And Gate		

Valves

	Hand-Operated Gate Valve		Check Valve 2		Solenoid Valve		4-way Plug Valve		Pinch Valve
	Gate Valve		Butterfly Valve		Hydraulic Valve		4-way Valve		Minimum Flow Valve
	Closed Gate Valve		Flanged Valve		Motor-Operated Valve		Electro-Hydraulic Valve		Auto Recirculation
	Hand-Operated Globe Valve		Flanged Valve 2		Pilot Gate Valve		Balanced Diaphragm Gate Valve		Gauge
	Globe Valve		Angle Valve Hand-Operated		Weight Gate Valve		Spring Gate Valve		Bleeder Valve
	Rotary Valve		Angle Valve		Slide Valve		Ram Valve		Integrated Block Valve
	Needle Valve		Relief Valve		Metering Coke		Maual Integrated Valve		Orifice
	Control Valve		Angle Valve		Knife Valve		Rotameter		Quarter Turn Valve Double Acting
	Piston-Operated Valve		Angle Blowdown Valve		Excess Flow Valve		Quarter Turn Valve Spring Acting		Water Flow Meter
	Back Pressure Regulator		Ball		Post Indicator		Self Draining Valve		Self-Operating Release Valve
	Plug or Cock Valve		Normally Closed Ball		Diaphragm Valve				
	Check Valve		Plug Valve						

Pumps

	Compressor		Axial Compressor		Sump pump		Reciprocating pump		Diesel motor
	Reciprocating Compressor		Centrifugal pump		Positive displacement pump 2		Turbine pump		ISO Liquid pump
	Reciprocating Compressor 2		Centrifugal pump 2		Rotary pump		Pump 2		ISO centrifugal pump
	Compressor silencers		Centrifugal pump 3		Rotary gear pump		Fan		ISO diaphragm pump
	Centrifugal		Centrifugal pump 4		Peristaltic pump		Selectable fan		ISO gear pump
	Rotary Compressor		Gas blower		Gear pump		Fan blades		ISO Progressive pump
	Rotary Compressor 2		Vertical pump		Proportioning pump		Fan blades 2		ISO positive displacement pump
	Rotary Compressor and silencers		Liquid ring vacuum pump		Vacuum pump		Triple fan blades		ISO reciprocating piston pump
	Liquid ring Compressor		Positive displacement pump		Screw pump		Axial flow fan		ISO screw pump
	Centrifugal Compressor		Horizontal pump		Screw pump 2		Turbine driver		Selectable Compressor
	Centrifugal Compressor 2		Vertical pump		Cavity pump		Doubleflow turbine		Pump
	Centrifugal Compressor 3				Submersible pump		Motor driven turbine		Positive displacement pump
	Centrifugal blower				Reciprocating pump		Ejector		Motor
							Spray		

Vessels

	Vertical Vessel		Packing column		Boiler		Covered Tank		Thermal Insulation Vessel
	Mixing Vessel		Drum		Dome Boiler		Floating Roof Tank		Heating-cooling Jacket Vessel
	Column		Knock-out drum		Hot Liquid Boiler		Open Bulk Storage		Brackets Vessel
	Tray Column		Bag		Tank		Dome Roof Tank		Dished Ends Vessel
	Fluidized Bed Column		Bag (ISO)		Tank 2		Cone Roof Tank		Pit Vessel
	Staggered Baffle Trays Column		Barrel		Tank 3		Internal Floating Roof Tank		Electrical Heating Vessel
	Packing Column		Barrel (ISO)		Tank 4		Double Wall Tank		Wastewater Treatment
			Gas bottle		Open tank		Onion Tank		Bin

Filters

	Mode Filter		Filter		Fixed bed air Filter		Biological Filter		High efficiency Filter
	Mode Filter indication		Filter 2		HEPA gas Filter		Ionexchanger Filter		Primary Filter Mid-effect Filter
	Filter		Gas Filter		Liquid Filter		Press Filter		Oxygen-enriched Filter
	Highpass Filter		Cartridge gas Filter		Fixed bed liquid Filter		Air Filter		Filter
	Lowpass Filter		Roll air Filter		Rotary liquid Filter		Air Filter 2		Filter 2
	Bandpass Filter						The Filter		Rotary Filter
									Suction Filter

Compressors

	Compressor		Ejector Compressor		Compressor silencers		Liquid ring Compressor		Axial Compressor
	Compressor 2		Piston Compressor		Rotary Compressor		Centrifugal Compressor		Air Compressor
	Compressor, Vacuum pump		Ring Compressor		Rotary Compressor 2		Centrifugal Compressor 2		AC air Compressor
	Centrifugal Compressor		Roller vane Compressor		Rotary Compressor 3		Centrifugal Compressor 3		Screw Compressor
	Diaphragm Compressor		Reciprocating Compressor 3		Rotary Compressor and silencers		Selectable Compressor		Turbo Compressor

Heat Exchanges

	TEMA TYPE BEM		Spray Cooler		Shell and Tube Heat 3		Reboiler		Oil Burner
	TEMA TYPE BEU		Forced-draft Cooling Tower		Cooler		Single Pass Heat Exchanger		Fired Heater
	TEMA TYPE AEM		Induced-draft Cooling Tower		Air-blown Cooler		Single Pass Heat Exchanger		Vertical Turbine
	TEMA TYPE AEL		Heat Exchanger		Induced Flow Air Cooler		Double Pipe Heat Exchanger		Condenser
	TEMA TYPE NEN		Heat Exchanger 2		Fin-fan Cooler		Hairpin Exchanger		Extractor Hood
	TEMA TYPE BKU		Heat Exchanger 3		Electric Heater		Spiral Heat Exchanger		Hose reel
	Plate Exchanger		Heater		Straight Tubes Heat Exchanger		Spiral Heat Exchanger 2		Light Water Station
	Plate and Frame Heat Exchanger		Exchanger		Coil Tubes Heat Exchanger		Air Cooled Exchanger		Combustion Chamber
	Dryer		Condenser		Finned Tubes Heat Exchanger		Briquetting Machine		Silencer
	Cooling Tower		Shell and Tube Heat		Floating Head Heat Exchanger		U-Tube Heat Exchanger		Thin-Film Evaporator
	Cooling Tower 2		Shell and Tube Heat 2		Plate Heat Exchanger		U-tube Heat Exchanger		Vent
	Cooling Tower 3				Kettle Heat Exchanger		Boiler		
					Reboiler Heat Exchanger				

Dryers

	Dryer		Roller Conveyor Belt Dryer		Generator		Heat Consumer		Moving Shelf Dryer
	Fluidized Bed Dryer		Motor Generator		Drying Oven		Spray Dryer		

General

	Crossflow Induced Draft		Plate Tower		Fluid Catalytic Cracking		Fluidized Reactor		Screen
	Counterflow Forced Draft		Packed Tower		Fluid Coking		Alkylation		Boom Loader
	Chimney Tower Hyperbolic		Hydrocracking		Tubular		Furnace		Automatic Stoker
	Furnace		Hydrosulfurization				Hydrocracking		Electric Motor
									Transport

Mixers

	In-line Mixer		Various Mixers		Agitator, Propeller		Agitator, Turbine		Agitator, Disc
	Mixer		Various Mixers 2		Agitator, Anchor		Agitator, Helical		Agitator, Cross-beam
	Various Mixers		Agitator, Stirrer		Agitator, Impeller		Agitator, Gat Paddle		

Crushers

	Crusher		Medium Crushers		Various Crushers		Jaw Crusher		Vibration Crusher
	Coarse Crusher		Fine Crushers		Hammer Crusher		Roller Crusher		Hammer Crusher
					Impact Crusher		Cone Crusher		

Centrifuges

	Centrifugal Pump		Centrifuge Screw		Centrifuge, Pusher		Centrifugal Blower		Centrifugal Pump 3
	Centrifugal Compressor		Centrifuge, Solid Shell		Centrifuge Screw, Perforated Shell		Centrifugal Pump		Centrifugal Pump 4
	Centrifugal Compressor 2		Centrifuge, High Speed		Centrifuge, Separator disc		Centrifugal Pump 2		Centrifugal Fan
	Centrifugal Compressor 3		Centrifuge, Perforated Shell		Centrifuge, Skimmer		Centrifuge		Centrifugal Pump 5

Motors

	Motor		AC Motor		Generator		Motor-operated Valve		Diesel Motor
	Step Motor		Motor Generator		AC Generator		Motor-driven Turbine		Electric Motor
	DC Motor		Gear		DC Generator		Motor		Turbine

Peripheral

	Impact Separator		Injector		Elevator		Elevator 2		Vertical Shaping Machines
	Gravity Separator		Reducer		Z-form Elevator		Skip Hoist		Pelletizing Disc
	Cyclone Separator		Rupture Disc		Lift		Conveyor		Piston Press
	Electromagnetic Separator		Viewing Glass		Hoists		Conveyor, Chain, Closed		Roller Press
	Electrostatic Precipitator Separator		Viewing Glass with Lighting		Aerator with Sparger		Conveyor, Screw, Closed		Crane
	Solidifier		Back Draft Damper		Bag Filling Machine		Roller Conveyor		Cyclone
	Permanent Magnet		Box Truck		Belt Skimmer		Conveyor		Curved Gas Vent
	Spray Nozzle		Manual Forklift		Screening		Overhead Conveyor		Firing System
	Rotary Table Feeder		Truck Forklift		Bucket Elevator		Screw Conveyor		Flame Arrestor
	Proportional Feeder		Industrial Truck		Elevator 1		Scraper Conveyor		Palletizer
	Rotary Valve Feeder		Rolling Bin Truck		Gas Flare		Piston Extruder Shaping Machines		Chimney
	Metering Proportional Feeder		Ship				Screw Extruder Shaping Machines		Conveyor, Vibrating, Closed

Piping and Connecting Shapes									
	Major Pipeline		Top to Top		Double Containment		Electrically Insulated		Expansion Joint
	Connect Pipeline		Sonic Signal		Flange		In-line Mixer		Hose
	Major Straight Line Pipe		Nuclear		End Caps		Separator		Flexible Hose
	Straight Line Pipe		Pneumatic Control		End Cap 2		Bursting Disc		Flow Indicator
	Battery Limit Line		Pneumatic Binary Signal Line		Breather		Flame Arrestor		Bell Mouth